

M3 — Live Oral Defense

Individual · Assessed live; no separate submission · Weight and date: see the syllabus

What M3 is

You present your group's **improved analysis**, then defend it **individually** in a live Q&A.

M1 was the replication. M2 was another group's critique of it. M3 is where you show what you did with that critique — and answer for it in person.

What you submit

Nothing. M3 is assessed live and has no separate submission.

That does not mean there is nothing to prepare. The analysis you present has to exist, and has to have been run.

Before the defense

Your **M1 analysis is the starting point**. Between M2 and M3 your group receives its critique and decides what to act on — but M2 is an *input*, not the boundary of what you may change. Peer review is one source of insight; your own further analysis, diagnostics and second thoughts are others.

The critique of *your* M1 is yours to act on — that is what it came to you for. What you saw inside the M1 *you reviewed* is different: you may not copy or adapt that group's code, text, figures, output, or distinctive implementation. If reviewing their work showed you something to reconsider in your own analysis, justify and implement the change yourself and disclose the influence in your cover note.

Three kinds of change are equally legitimate at M3:

1. **Reviewer-prompted** — “M2 raised this, so we tested it / changed it.”
2. **Self-initiated** — “Revisiting the analysis ourselves, we found this and changed it.” No reviewer has to have mentioned it.
3. **No change, after investigation** — “We examined this and kept the original decision, because the sensitivity analysis did not alter the conclusion.” This is a real result, not an absence of work.

The requirement is **traceability, not obedience**. Any change to the analytic population, estimand, adjustment set, study design, variable definitions, missing-data strategy or model specification must be identified and justified — whoever prompted it. You are not obliged to implement a recommendation; you are obliged to answer it, and saying why you rejected it is a complete answer. Silently ignoring it is not.

Whatever you did decide to act on, **run it**. The estimate you defend must come from code that executed, not from a change you intended to make. You will be asked what it produced.

What the defense is about

The presentation shows *what* you did. The defense is about **whether it was justified**:

- which reviewer points you **accepted**, and which you **rejected** — and on what grounds;
- what you changed **that no reviewer asked for**, and what led you to it;
- whether the change addresses a **real methodological threat**, or a cosmetic one;
- **what the change did to the estimate** — did it move, by how much, and did the uncertainty around it change too;
- whether that **changes, or fails to change, the interpretation** — in a way you can explain and stand behind.

“We made the change and it made no difference to the conclusion” is a good answer when you can show it. A change that moves nothing is still evidence about how robust the original finding was.

Know your own numbers

Expect to be asked about the analysis itself, not only the reasoning around it:

- what the estimate is now, and what it was before the change;
- which quantity you are actually estimating, and whether it is the one the claim needs;
- how the study design, the weighting, or the missing-data handling entered the result you are showing;
- where a number on your slide came from in your code.

You are not graded on whether your code merely runs — that is assumed by this point in the course. You are graded on whether you can account for what it produced.

During the defense

In your group’s presentation, use everything you prepared — your slides, your own speaker notes, your own code and output.

In individual Q&A, the answer has to be yours, spoken. Look at your own slides and your own code and output as much as you need. Do not read from a prepared answer, do not take help from anyone in any channel, and do not consult anything produced during the session — no AI, no live prompting, no messages, no transcription or notetaking tool.

Approved accommodations are the exception. A support authorized by your Academic Accommodation Letter — a note-taker, live captioning, a scribe, permitted assistive software — is arranged with the instructor beforehand and applies at M3 as it does anywhere else in the course. Nothing has to be raised or explained in the session itself; see the syllabus section *Academic Accommodation Letter*.

Prepare however you like, including rehearsing questions with an AI. There is nothing to disclose: M3 has no submission. AI that touched the code or slides you present is disclosed with M1 or M4.

This is an individual grade

M3 is the one milestone assessed per person rather than per group, so expect to be asked **which part of this was yours**. Every student is asked, so nobody is disadvantaged by not being asked.

Two things worth knowing in advance:

- **Conceding a point counts in your favour.** A student who says “that is a fair criticism, and here is what I would do about it” is demonstrating exactly what is being assessed.
- **Saying plainly where an answer is not known is a strength**, provided you can say what it would take to find out. “I don’t know” on its own is not; “we did not test that, and it would need X” is.

How it is graded

The usual three-band scale applies:

- **Exceptional** — engages with follow-ups that go beyond the presented work, including the limits of what is known.
- **Adequate** — answers follow-ups that stay within the group’s presented work.
- **Weak** — answers only what is on the slides.

Weight, deadline and policy

See the syllabus. It governs what this milestone is worth, when it is due, whether the weight can be shifted, and any late penalty. This page describes the task only.